

**CORRELATIVE STUDY FOR BIOCHEMICAL COMPOUNDS OF
FLOWER-BASED EXTRACT FROM *COUROUPITA GUIANENSIS*: GC-
MS CHARACTERIZATION, ANTIMICROBIAL EFFICACY,
INHIBITION OF PROTEIN DENATURATION ACTIVITY, AND BRINE
SHRIMP LETHALITY TEST**

A Dissertation submitted to
Bishop Heber College (Autonomous)
(Affiliated to Bharathidasan University, Tiruchirappalli)
In partial fulfilment to the requirements for the award of the degree of

BACHELOR OF SCIENCE IN BIOTECHNOLOGY

Submitted By

SIVABALAN N	235117240
SNEKHA R	235117241
SRI ASHWATHI T	235117243
VAISHNAVI PERUMAL YASOTHA	235117249

Under the guidance of

Ms A. AMRITA ANANDI
Assistant Professor,
Department of Biotechnology,
Bishop Heber College (Autonomous),
Tiruchirappalli-17



DEPARTMENT OF BIOTECHNOLOGY
BISHOP HEBER COLLEGE (AUTONOMOUS)

Affiliated to Bharathidasan University
(Nationally Reaccredited with 'A++' Grade by NAAC with a CGPA of 3.69 out of 4)
(Recognized by UGC as "College of Excellence")
(Star College status, awarded by the DBT|DST-FIST sponsored college)
Tiruchirappalli-620017

OCTOBER-2025



DEPARTMENT OF BIOTECHNOLOGY
BISHOP HEBER COLLEGE (AUTONOMOUS)

Affiliated to Bharathidasan University

Nationally Reaccredited with 'A++' grade by NAAC with a CGPA of 3.69 out of 4

Recognized by UGC as "College of Excellence"

Star College status, awarded by the DBT|DST-FIST sponsored college

Tiruchirappalli-620017

OCTOBER-2025

CERTIFICATE

The viva-voce examinations for the candidates,

SIVABALAN N	235117240
SNEKHA R	235117241
SRI ASHWATHI T	235117243
VAISHNAVI PERUMAL YASOTHA	235117249

was held at Bishop Heber College (Autonomous), Tiruchirappalli – 17.

EXAMINERS:

1. Internal:

Signature.....

Name.....

2. External:

Signature.....

Name.....

SIGNATURE OF THE HOD



Ms A. AMRITA ANANDI
Assistant Professor,
Department of Biotechnology,
Bishop Heber College (Autonomous),
Tiruchirappalli–620017

DATE:

CERTIFICATE

This is to certify that the dissertation entitled “Correlative study for Biochemical Compounds of Flower-Based Extract from *Couroupita guianensis*: GC-MS Characterization, Antimicrobial Efficacy, Inhibition of Protein Denaturation Activity, and Brine Shrimp Lethality Test” is the bonafide work done under my guidance by SIVABALAN N (235117240), SNEKHA R (235117241), SRI ASHWATHI T (235117243), VAISHNAVI PERUMAL YASOTHA (235117249) and submitted to Bishop Heber College (Autonomous) Tiruchirappalli – 17, in a partial fulfilment of the requirement for the award of the degree of Bachelor of Science in Biotechnology during the academic year (2023 – 2026) and has not been included in any thesis or dissertation previously for any degree.

Signature of the HOD

Signature of the Internal Guide

DECLARATION

We hereby declare that the work presented in this dissertation is the original work done by us under the guidance of **Ms A. AMRITA ANANDI, M.Sc., NET, SET, GATE, Assistant Professor**, Department of Biotechnology, Bishop Heber College (Autonomous), Tiruchirappalli– 17 and has not been included in any other thesis/dissertation submitted for any other degree.

NAME OF THE STUDENT	ROLL NUMBER	SIGNATURE
SIVABALAN N	235117240	
SNEKHA R	235117241	
SRI ASHWATHI T	235117243	
VAISHNAVI PERUMAL YASOTHA	235117249	

ACKNOWLEDGEMENT

First of all, we praise God, the Almighty, who is merciful and passionate for showing his blessings to complete this project successfully. We would like to express our extreme gratitude to our honourable **PRINCIPAL, Dr J. PRINCY MERLIN**, Bishop Heber College (Autonomous), Tiruchirappalli, for permitting us to do our project work at our institution.

We express our gratitude to **Dr R. JASMINE JEMIMAH JEYANTHI, HEAD OF THE DEPARTMENT**, Department of Biotechnology and Bioinformatics, Bishop Heber College (Autonomous), Tiruchirappalli, for her support in completing my project successfully.

We would like to thank **Ms A. AMRITA ANANDI, M.Sc., NET, SET, GATE, Assistant Professor**, Department of Biotechnology, for her understanding, guidance, and encouragement. We wish to place our thanks to all the staff members of the Department of Biotechnology, Bishop Heber College (Autonomous), Tiruchirappalli, for their encouragement and support. We would like to thank our parents and our guide for their unlimited affection and care for us.

Finally, we would also like to thank all our class friends who had helped us in various ways in the completion of the work.

CONTENTS

SERIAL NUMBER	TITLE	PAGE NUMBER
	ABSTRACT	
1.	INTRODUCTION	
2.	REVIEW OF LITERATURE	
3.	MATERIALS AND METHODS	
4.	RESULTS	
5.	DISCUSSION	
	CONCLUSION	

Correlative study for Biochemical Compounds of Flower-Based Extract from *Couroupita guianensis*: GC-MS Characterization, Antimicrobial Efficacy, Inhibition of Protein Denaturation Activity, and Brine Shrimp Lethality Test.

ABSTRACT

Couroupita guianensis, commonly called the Cannonball tree, is a rare and culturally significant member of the Lecythidaceae family. Its flowers and fruits have drawn attention not only for their ornamental value but also for their traditional medicinal applications. Across several medical systems, particularly Ayurveda and Chinese medicine, the plant has been used as a remedy for conditions such as fever, inflammation, infections, pain, ulcers, and even tumors. Despite these widespread uses, systematic scientific evaluation of its phytochemical composition and pharmacological potential remains limited.

The present investigation focused on assessing extracts of *C. guianensis* obtained with different solvents (ethanol, acetone, chloroform, and water). Each extract was evaluated for antibacterial, antioxidant, anti-inflammatory, and cytotoxic activity, and chemical composition was profiled using standard phytochemical tests and gas chromatography–mass spectrometry (GC–MS). Additionally, molecular docking was employed to predict the binding efficiency of bioactive compounds with target proteins associated with oxidative stress and inflammatory pathways.

Ethanolic extracts consistently exhibited stronger activity in antibacterial and antioxidant assays compared to chloroform and aqueous extracts, while acetone showed intermediate effects. GC–MS analysis revealed diverse phytoconstituents, several with documented biological relevance. Docking studies supported the experimental results, highlighting stable interactions of key metabolites with proteins such as *Protein 3 Inflammasome*, *Keap1* and *NADPH oxidase*. Overall, the study tries to establish the therapeutic potential of *C. guianensis* and provide a basis for advancing it toward preclinical and translational research under phytopharmacognosy.

Keywords: *Couroupita guianensis*, phytochemical, ethnomedicine, *in-silico*

1. INTRODUCTION

Plants are a crucial source of natural products. Almost every part of a plant has compounds that are active and can be used for treating illnesses. These plants usually provide a wide range of medicines, including antibiotics, antifungals, antiseptics, and pain relievers (Pandurangan et.al, 2018). For the last twenty years, plants in traditional medicine have caught the interest of scientists and drug makers because they work well against bacteria that are resistant to many medicines, often called superbugs. These medicines, due to high dosage and unsupervised recurrent usage, have caused a few side effects, which are especially harmful for immunocompromised individuals. The World Health Organisation says that antimicrobial resistance, or superbugs, are stated as a huge crisis in global health. In 2019, bacterial resistance caused 1.27 million deaths directly, and many more deaths were related to this issue. Without proper measures, the number of deaths from antibiotic resistance for infectious diseases could go up by 67%, reaching nearly 2 million every year by 2050. This might lead to around 39 million deaths in the next 25 years. The WHO also notes that 80% of people in South Asian and African countries use herbal medicines (Pandurangan et.al, 2018); which brings together traditional medical practices and modern science to find new ways to reintroduce ethnomedicinal plants for therapeutic drug development.

Scientific name: *Couroupita guianensis*

Common name: Cannonball tree

Conservation Status: IUCN (Lower Risk) Least Concern

General Description: A deciduous tree in the flowering plant family of Lecythidaceae, native to lowland tropical forests of Central and South America, with cultural, religious and medicinal significance in South and Southeast Asia.

Couroupita guianensis (Cannonball tree) is an evergreen species native to tropical regions, notable for its large woody fruits and distinctive blossoms. Its name derives from the resemblance of the fruit to iron cannonballs, which hang in clusters from thick stalks. In South Asia, the tree is often cultivated in temple courtyards, where the flower's unique shape is symbolically associated with the *Shivalingam* encircled by a serpent, giving it a strong cultural and religious identity. These traditional associations have helped preserve the species in sacred landscapes, even as it faces conservation concerns in natural habitats.

Beyond its symbolic significance, the plant has a long history of use in indigenous medicine. Traditional healers employ different parts of the tree—leaves, flowers, bark, and fruit, for managing diverse ailments. Reported therapeutic applications include treatment of skin disorders, microbial infections, gastric disturbances, inflammatory conditions, wounds, and pain. Some traditions also attribute anticancer and antiparasitic activity to the plant. Contemporary pharmacological investigations have substantiated numerous ethnomedicinal assertions, revealing antimicrobial, anti-inflammatory, antioxidant, and analgesic characteristics in experimental frameworks.

Nevertheless, research on *C. guianensis* remains fragmented. Most existing studies emphasise preliminary screenings or ethnobotanical reports, with fewer attempts to systematically analyse the phytochemical composition or investigate mechanisms of action. This creates a scientific gap between traditional knowledge and experimental evidence. Addressing this gap, the current study examined solvent extracts of *C. guianensis* using both experimental assays and computational approaches. The work involved phytochemical characterisation, antibacterial and antioxidant assessments, and evaluation of cytotoxicity. GC–MS was used to spot the active compounds, and molecular docking checked how they connect with proteins involved in free radical oxidative stress and inflammation. Together, these approaches aimed to generate a more comprehensive understanding of the pharmacological potential of this underexplored plant.

2. REVIEW OF LITERATURE

Couroupita guianensis, commonly known as the cannonball tree, is a member of the *Lecythidaceae* family with extensive ethnomedicinal applications across multiple plant parts. The plant demonstrates diverse pharmacological activities, including antibiotic, antiseptic, anti-inflammatory, antimicrobial, antimycobacterial, analgesic, antiarthritic, antibiofilm, antidiarrheal, antifertility, antipyretic, antistress, antitumor, antiulcer, antidermatophytic, wound healing, and immunomodulatory properties. Key bioactive compounds include isatin, tryptanthrin, indirubin, couroupitone A and B, steroids, glycosides, terpenoids, saponins, alkaloids, and phenolic substances (Sheba & Anuradha, 2020; Sundararajan & Koduru, 2014). Traditional uses encompass treating various ailments using leaves, flowers, fruits, and stems (Gousia et al., 2013). The plant shows promise for anti-cancer, anti-inflammatory, and antioxidant applications, though further in vivo research is needed to validate these traditional uses (Gousia et al., 2013). Research on medicinal plant use patterns within the *Lecythidaceae* family and broader contexts reveals significant ecological and cultural influences.

The *Lecythidaceae* family, comprising 25 genera and 400 species distributed pantropically, demonstrates extensive ethnopharmacological applications with 180 different metabolites reported from 25 species, predominantly triterpenes and flavonoids (Ferreira et al., 2021). Broader patterns show that ecosystem type significantly influences medicinal plant selection, with native and woody species dominating in Amazon, Cerrado, and Caatinga ecosystems, while this pattern differs in Atlantic Forest and Pampas regions (Medeiros et al., 2013). Statistical analyses reveal that botanical pharmacopoeias are non-random subsets of local floras, with only specific families showing significant departure from random selection patterns (Bennett & Husby, 2008). In Brazil's Caatinga vegetation, 56.25% of medicinal plants are native species, with usage patterns reflecting seasonal resource availability and supporting hypotheses of knowledge diversification rather than erosion (Albuquerque, 2006).

The methanol extract of *Couroupita guianensis* leaves showed antidiabetic activity against clinical pathogens. The plant extracts contain various phytochemicals such as carbohydrates, proteins, alkaloids, and phenolic compounds. *Couroupita guianensis* is identified as a medicinal plant with diverse pharmacological uses, including antidiabetic properties (Swapnalatha and Rajeswari, 2014).

In the Amazon rainforest, the shamans of the Mayantuyacu site use the healing virtues of decoctions and teas from different parts of *the Couroupita guianensis* Aubl. (*Lecythidaceae*) trees as remedies in Ashaninka medicine. However, the composition of the remedy and the underlying mechanism remain unclear. This study was designed to compare the metabolite profile of *Couroupita guianensis* bark decoction produced by Amazonian shamans with that obtained under standardised laboratory conditions and to investigate the biological properties of both decoction and isolated constituents in the skin wound healing process and inflammation. The chemical analyses were carried out by Ultra-High-Performance Liquid Chromatography coupled with UV and High-Resolution Mass Spectrometry detectors (UHPLC-UV-HRMS). 1D and 2D-NMR experiments were performed to identify the main decoction constituents. The decoction and pure compound effect on keratinocyte migration was determined by the in vitro wound healing model; the mechanism of action was elucidated by western blot analysis, which revealed the occurrence of polyphenolic compounds as catechins, ellagitannins and, notably, of unusual sulphated derivatives of ellagic acid isolated for the first time from *Couroupita guianensis* bark. A new natural sulphated molecule [4-(2''-O-sulphate- β -D-glucuronopyranosyl) ellagic acid] was identified as the potential active compound responsible for the efficacy of bark decoction stimulating wound healing in human HaCaT keratinocytes. The molecular mechanism involved the induction of pro-migratory pathways mediated by ERK and AKT phosphorylation and the increase of MMP2 expression in HaCaT cells. At the same time, the treatment inhibited inflammation, interfering with NF κ B activation.

Beyond identifying a new bioactive compound, the overall results scientifically validate the traditional use of *Couroupita guianensis* bark decoction as an anti-inflammatory remedy. Moreover, the beneficial effects on keratinocytes suggest promising therapeutic applications in skin diseases. This study thus identified a new sulphated derivative of ellagic acid in *Couroupita guianensis* bark decoction, which stimulates wound healing in human keratinocytes.

The molecular mechanism involves the induction of pro-migratory pathways and the inhibition of inflammation. The findings validate the traditional use of the bark decoction as an anti-inflammatory remedy and suggest potential therapeutic applications in skin diseases. (Esposito et al. 2023)

Couroupita guianensis flower extracts exhibit notable antibacterial and antifungal activities, attributed to their rich phytochemical content, including alkaloids, terpenoids, phenolic compounds, flavonoids, and steroids (C. Ramalakshmi et al., 2013; Pusapati Madan Ranjit & G. Guntuku, 2014; Reetika Singh et al., 2015). Methanol and other organic extracts of the flowers demonstrated effective inhibition against pathogens such as *Staphylococcus aureus*, *Plesiomonas shigelloides*, *Vibrio mimicus*, *Proteus vulgaris*, and moderate activity against *Escherichia coli*, *Salmonella typhi*, and *Klebsiella pneumoniae* (C. Ramalakshmi et al., 2013; Pusapati Madan Ranjit & G. Guntuku, 2014). Both flower and leaf extracts showed antifungal activity, with leaf extracts generally exhibiting higher potency (H. Raghavendra et al., 2017; Reetika Singh et al., 2015). The presence of alkaloids is particularly associated with antimicrobial effects (C. Ramalakshmi et al., 2013). These findings suggest *C. guianensis* flowers as a promising source for developing new antimicrobial agents, though further isolation and characterisation of active compounds are needed (Pusapati Madan Ranjit & G. Guntuku, 2014; Reetika Singh et al., 2015).

Research on *Couroupita guianensis* demonstrates significant anti-inflammatory and cytotoxic properties comparable to standard pharmaceutical interventions. *C. guianensis* seed oil contains tetranortriterpenoids, including gedunin and andirobin, which exhibit marked anti-inflammatory effects by inhibiting edema formation and reducing inflammatory mediators like TNF- α and IL-1 β through NF κ B pathway inhibition (Henriques & Penido, 2014). These compounds also demonstrate anti-allergic properties and cellular activation inhibition in mast cells, eosinophils, and T lymphocytes (Machado & Mendes, 2021).

Couroupita guianensis extracts show comparable efficacy to reference drugs, with doses of 30-100 mg/kg significantly reducing inflammatory responses in experimental models, nearly abolishing leukocyte migration at higher doses like standard treatments (Pinheiro et al., 2013). The bark decoction contains novel sulphated ellagic acid derivatives that promote wound healing through ERK and AKT phosphorylation while inhibiting NFκB-mediated inflammation, validating traditional anti-inflammatory applications (Esposito et al., 2023).

Molecular docking studies have identified several potential therapeutic targets and binding affinities for compounds from *Couroupita guianensis* and related phytochemicals. Compounds from *C. guianensis* exhibited favourable binding energies and hydrogen bond interactions with selected protein receptors, showing good drug-likeness properties and pharmacokinetic profiles through computational evaluation using Argus Lab and other in silico tools (Kanerla et al. 2019). Plant-derived compounds, including scopodulcic acid and dammar enolic acid, showed strong binding affinities to SARS-CoV-2 targets (spike glycoprotein and Mpro protein), with silybinin demonstrating multitarget potential (Ubani et al. 2020).

Couroupita guianensis, commonly known as the cannonball tree, is a member of the Lecythidaceae family native to tropical northern South America and the Southern Caribbean (Kumar et al., 2011). The plant has extensive traditional medicinal applications, with almost all parts, including leaves, flowers, fruits, and stems, used to treat various ailments (Sheba & Anuradha, 2020; Gousia et al., 2013). Traditional uses encompass antipyretic, antidepressant, antiseptic, analgesic, anti-inflammatory, antiprotozoal, anticancer, antiulcer, antimicrobial, wound healing, antidiarrheal, antioxidant, and antifertility activities (Sundararajan & Koduru, 2014). The plant contains numerous bioactive compounds, including steroids, glycosides, couroupitone A and B, isatin, triterpenoids, eugenol, tryptanthrin, and phenolic substances (Sundararajan & Koduru, 2014; Sheba & Anuradha, 2020). Despite its widespread traditional use, the plant remains relatively unexploited from an ethnopharmacological perspective and requires further investigation to isolate pharmacologically active compounds for novel drug development (Kumar et al., 2011; Gousia et al., 2013).

3. MATERIALS AND METHODS

3.1 Plant material collection:

Couroupita guianensis flowers were freshly collected in July 2025 from the Bishop Heber College campus botanical garden in Trichy and Jeeyapuram, Trichy, Tamil Nadu, India. The flowers collected were hand-picked using sterile gloves and washed with distilled water to exclude dust particles. The collected flowers were then dehydrated and powdered using a mechanical grinder. The nature of the powder was identified depending on the characteristics such as colour, odour, taste and texture, which were examined physically and are listed below:

Colour: Light Brown-Green

Odour: Earthy-Pungent Smell

Taste: Astringent

Texture: Gritty-Chalk-like texture



Figure 3.1.1



Figure 3.1.2



Figure 3.1.3

3.2 Cold Extraction:

The sample (40 g) of the collected herbal sample was dissolved in chloroform, acetone, and ethanol, followed by the orbital shaker method for three continuous days. Later, the extracts were removed free of particulate matter using Whatman filter paper. The solvents were safely covered and stored in sterile Blue Screw Cap Bottles.



Figure 3.2.1



Figure 3.2.2

3.3 Soxhlet Apparatus for Aqueous Hot Extraction:

The process begins with assembling the Soxhlet apparatus using laboratory instruments to set up the extraction apparatus. Once the setup is in place, 300 mL of ethanol or distilled water is measured and poured into a round-bottom flask, which is then attached to the extractor and condenser, positioned accordingly. Next, 30 g of the sample is placed into a thimble. The arm of the extractor is covered with glass wool to prevent heat loss. Heating causes the solvent to evaporate and travel up through the apparatus to the condenser. The vapour turns back into liquid and drops into the extraction chamber. As the solvent level rises and reaches the top of the siphon, it automatically drains back into the round-bottom glassware, completing one extraction cycle. This process continued in a loop and typically ran for 16 hours.

Once the extraction is underway, it can operate without constant supervision. However, due to the combination of running water (in the condenser) and an electrical heating device, it is not recommended to leave the apparatus completely unattended. The extract is set aside to cool down.

3.4 Rotary evaporator:

The principle behind rotary evaporators is that solvents have a range of boiling temperatures, and that range decreases with decreasing pressure. A large region of thin film forms on the interior surface of the evaporation flask; as a result, the materials are forced to rotate at a default speed setting. For example, the temperatures were adjusted corresponding to their boiling points, i.e. acetone was set to 46°C, chloroform was set to 51°C, and ethanol was set to 68°C.

Materials with a lower melting point evaporate quickly when the flask is heated uniformly. This method of extraction produced higher extract yields.



Figure 3.4.1

3.5 Preliminary Phytochemical Screening:

The extracts were subjected to qualitative phytochemical tests to detect the presence of different chemical groups of compounds. Organic plant concentrates stored previously after being rotary evaporated were screened for the presence of lipids, saponins, tannins, alkaloids, flavonoids, triterpenes, sterols, glycosides, quinones, coumarin, gum, mucilage, carbohydrates, reducing sugars, starch, protein, and amino acids, as described in the literature.

Procedure for Qualitative Phytochemical Screening:

Test for Alkaloids:

1. Wagner's test: 2 ml of extract was treated with a few drops of Wagner's reagent. Formation of reddish-brown precipitate indicates the presence of alkaloids.

Test for Flavonoids:

1. Alkaline test: 2 ml of extract was added to a few drops of NaOH solution. Formation of intense yellow colour, which turned colourless on addition of a few drops of diluted HCl, indicates the presence of flavanols.

Test for Proteins:

1. Biuret test: 2 ml of extract was added to 2 ml of 5% NaOH and 2 ml of 1% CuSO₄ solutions. Violet or Purple colouration confirms the presence of proteins.
2. Xanthoprotein test: 2 ml of extract was treated with a few drops of concentrated nitric acid. Formation of yellow colour indicates the presence of proteins.
3. Concentrated H₂SO₄ test: 2 ml extract was treated with a few drops of conc. H₂SO₄. Formation of milky precipitate confirms the presence of proteins.

Test for Saponins:

1. Foam Test: 2ml of extract was diluted with 10ml of distilled water and warmed gently. It was shaken for 15 minutes continuously. Persistent froth indicates the presence of saponins.

Test for Tannins:

1. HCl Test: 2ml of extract was added to dilute HCl. Formation of Red Precipitate indicates the presence of tannins.
2. Braymer's Test: To 2mL of extract, 1mL of 5% FeCl₃ solution was added along with a few drops of H₂O. Formation of green precipitate indicates the presence of tannins.

Test for Coumarins:

2ml of extract was added to 3ml of 10% NaOH solution. Yellow colouration indicates the presence of coumarins.

Test for Quinones:

1. Alcoholic KOH Test: To 2ml of plant extract, alcoholic KOH was added. Formation of red-blue colour indicates the presence of quinones.

Test for Glycosides:

2mL of extract was treated with 1 mL of water and shaken well. To this mixture, aqueous NaOH was added. Formation of a yellow precipitate indicates the presence of glycosides.

Test for Sterols and Terpenoids:

1.Salkowski's Test: To 1mL of extract, 2mL of chloroform and 1mL of concentrated H₂SO₄ were added. Reddish-Brown colouration indicates the presence of sterols and terpenoids.

Test for Reducing Sugars:

1.Fehling's Test: 2mL of extract was hydrolysed with diluted HCl, neutralized with alkali and heated with Fehling's A and B solutions. Formation of red precipitate indicates the presence of reducing sugars.

2.Benedict's test: 2 ml of extract was added to 1ml of Benedict's reagent and heated gently. Orange-red precipitate indicates the presence of reducing sugars.

Test for Lipids:

1.CuSO₄ test: 2 ml of extract added to 1 ml of 1% CuSO₄ solution and 10% NaOH solution. Blue colouration indicates the presence of fixed oils.

Test for Phenols:

1. Ferric chloride test: 2 ml of extract was treated with 2-3 drops of 5% FeCl₃ solution. Formation of bluish-black colour indicates the presence of phenols.

2. Lead Acetate test: 2ml of extract was dissolved in distilled water, and to this, 3 mL of 10% lead acetate solution was added. A chalk-like white precipitate indicates the presence of phenols.

Test for Triterpenes:

Add 2ml of plant extract with 2ml of chloroform, and add 2ml of Acetic Acid and a few drops of Concentrated H₂SO₄. Deep red junction will form.

Test for Carboxylic Acid:

1. Sodium Bicarbonate test: Add 2mL of extract with 4ml of NaHCO₃ solution. Effervescence indicates the presence of carboxylic acids.

3.6 GC-MS Analysis:

Gas Chromatography-Mass Spectrometry (GC-MS) represents an analytical technique that merges the capabilities of Gas Chromatography (GC) with Mass Spectrometry (MS). Through internal standards, this method offers comprehensive capabilities for identifying and quantifying compounds in analysed materials (Skoog et al., 2007). Gas chromatography functions as a chromatographic analytical technique where the stationary phase consists of a thin liquid or polymer layer contained within glass or metal tubing, while the mobile phase typically comprises an inert gas such as helium or an unreactive gas like nitrogen. In this system, mass spectrometry serves as the detection mechanism for gas chromatography.

While Gas Chromatography demonstrates exceptional separation capabilities with outstanding resolution, it cannot directly identify volatile or semi-volatile compounds. However, Mass Spectrometry compensates for this limitation by delivering comprehensive structural information that enables accurate identification of most chemical compounds.

Specification and Procedure:

Flow gas: Helium

Instrument: SHIMADZU

Column model: SH-RXU-5Sil MS

The ethanolic extract obtained from the extraction was subjected to GC–MS analysis to identify the phytochemicals present in the flower extract of *C. guianensis* at the HAIF facility, Bishop Heber College.

The GC–MS examination was performed using a SHIMADZU/QP2020GC setup that included an AOC–20i autosampler and gas chromatograph connected to a mass spectrometer (GC-MS) unit under the specified parameters: injection flow rate of 1.2 mL/min, injector temperature maintained at 250 °C, and ion-source temperature held at 200 °C.

Mass spectral data were acquired at 0.3-second scanning intervals, capturing fragments ranging from 50 to 500 Da. The interpretation of GC–MS mass spectra was conducted utilising the WILEY8LIB database from the online Wiley library. Unknown compound mass spectra were matched against reference spectra contained within the WILEY8LIB database. Component identification, including compound names, molecular weights, and structural information of the test samples, was determined and organised in tabular format within the results section.

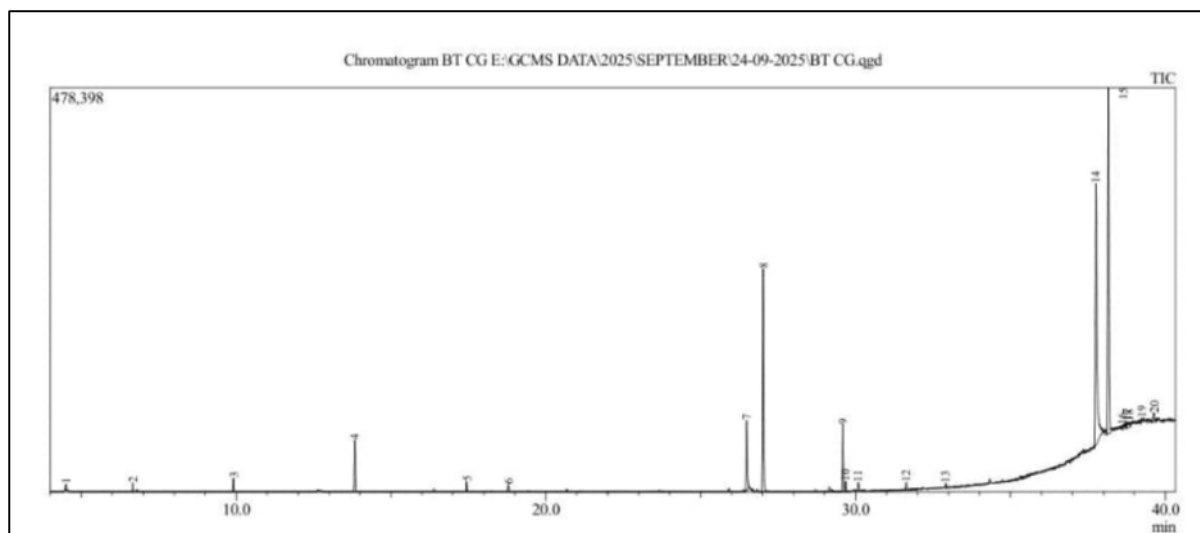


Figure 3.6.1

3.7 ANTIMICROBIAL EFFICACY ASSAY:

Materials and Glassware:

Beakers, Conical Flasks, Petri Plates, Agar-Agar, Muller-Hinton Agar (MHA), Rose-Bengal Agar (RB Agar), LB Broth, Test tubes, Laminar Airflow Chamber, Sterile PPEs, L rods, Micropipettes and other essential materials for assay.

Samples:

C. guianensis flower concentrated samples, ethanol, acetone and chloroform were mixed with DMSO solution for a diluted sample.

3.7.1 ANTIBACTERIAL ASSAY:

Sterilization:

Sterilization was completed using a pressure cooker, which involves securely covering glassware and agar and placing them in the pressure cooker. It is essential to sustain a temperature of 121°C for 15 to 20 minutes at a pressure of 15 PSI (pounds per square inch). This method demonstrates that sterilization via a pressure cooker can be efficiently performed in rural healthcare settings or at home. This technique is superior to boiling water as steam under pressure reaches a higher temperature. Furthermore, it reduces the risk of contaminating cotton-wrapped materials due to handling after sterilization.

Sub-culture and storage of bacterial cultures:

The bacterial inocula, such as *Streptococcus pneumoniae*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, and *Pseudomonas aeruginosa*, were grown in LB broth in the Department of Biotechnology PG Laboratory at Bishop Heber College in Trichy. The bacteria were cultured on Muller-Hinton agar, perfect for testing antimicrobial activity, and the cultures were incubated at 37°C for a whole day.

Gram staining:

Gram staining represents a fundamental technique in microbiology for bacterial identification. This method was developed in 1882 by Hans Christian Gram, a Danish bacteriologist, who initially used it to detect pathogens responsible for pneumonia. The procedure typically serves as the primary diagnostic test performed in laboratory settings. The technique employs crystal violet or methylene blue as the primary cationic dye. Bacterial specimens that successfully retain this dye display a purple-brown colouration when examined under microscopic observation and receive classification as Gram-positive organisms. Conversely, bacterial samples that fail to hold the cationic dye exhibit a red appearance and are designated as Gram-negative bacteria. The effectiveness of Gram staining depends on the interaction between the dye and various bacterial species, which is determined by the distinct physical and chemical characteristics of their cellular wall structures. Under standardized laboratory protocols, researchers prepare bacterial culture smears on glass microscope slides, then systematically apply the necessary sequence of staining agents. The Gram-staining technique was implemented to confirm each bacterial culture.

Procedure:

- Apply crystal violet stain to the fixture on the glass slide.
- After 10 to 60 seconds, remove the stain and rinse off any excess with distilled water.
- Apply Gram's iodine solution to the smear for 1 minute. Discard the iodine solution and rinse the slide with running water.
- Add a few drops of decolouriser (ethanol) by slightly tilting glass slide, rinse with water to prevent over-colourisation. Stop adding decolouriser as soon as the solvent appears colourless while flowing over the slide.
- Finally, counterstain the smear with safranin solution for 3 minutes, then remove excess stain using sterile distilled water.
- The slide can be air-dried after gently shaking off excess water.
- Examine it under a microscope.

Well Diffusion Technique:

Antibacterial test was completed using the well diffusion technique. Muller-Hinton agar was prepared and sterilized. The sterile petri dishes were autoclaved for 15 minutes at 121°C. The medium was poured into the petri dishes following aseptic procedures (within a laminar air flow chamber) and allowed to solidify. Once the medium was solid, the test cultures were applied to the agar plates using the spread plate method. Wells were made in the medium to introduce the flower extracts (sample). The examination was carried out with the use of three distinct concentrations of *C. guianensis* flower extracts (100µl, 50µl, 25µl) placed in the wells. The plates were incubated at 37.5°C for a day.

Procedure:

The required volume (20 -30 ml) of the sterilized Muller-Hinton agar media was poured onto sterilized Petri dishes and then allowed to solidify. The 100 µL standardised inoculum was inoculated with the help of an L-shaped glass rod onto the agar plates. The wells were loaded with varied concentrations of extracts made previously and were compared with a positive control of an antibiotic tablet, Gentamycin and a negative control- distilled water. The plates were set aside in a controlled chamber at 37.5°C for a full day, and the susceptibility was determined based on the inhibition zones. The diameter of the zones of inhibition was noted down in terms of millimetres using a ruler. The experiments were repeated in three replicates, and the mean values are presented as results along with their standard deviations. Standard deviations of the positive control (Gentamycin) were also calculated and are presented in the same table for comparative study. Results were noted down as the zone of inhibition is usually measured in millimetres. This process was repeated three times to validate the test findings.

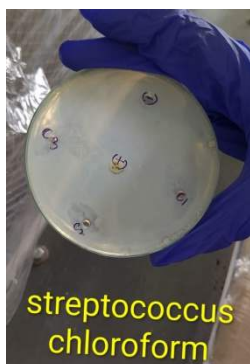


Figure 3.7.1.1



Figure 3.7.1.2



Figure 3.7.1.3

3.8 ANTI-INFLAMMATORY ASSAY:

Inhibition of Bovine Serum Albumin (BSA) Protein Denaturation Activity:

Preparation of 3% BSA solution: 30mg of Bovine Serum Albumin (BSA) was added to 80mL of sterile distilled water and made up to 100mL.

To assess the anti-inflammatory activity qualitatively, a reaction mixture of 5 mL was prepared by combining 0.2 mL of 3% BSA solution, 2 mL of the aqueous sample extract and standard (Aspirin) at varying concentrations, with 2.8 mL of phosphate-buffered saline (pH 7.4). A control of the same volume was made by combining 2 mL of triple-distilled water, 0.2 mL of 3% BSA solution, and 2.8 mL of phosphate-buffered saline (pH 7.4). The reaction mixture was left at room temperature for 30 minutes, then heated in a water bath at 72°C for 15 minutes. After cooling, absorbance was measured at 280 nm using a UV/Vis spectrophotometer with triple-distilled water as the blank. The samples were then incubated at room temperature for 25 minutes and heated at 55°C for 30 minutes on a water bath before cooling again for absorbance assessment at the same wavelength. Aspirin was used as a standard for comparison, with dimethyl sulfoxide (DMSO) as a control in this study.

The proportion of protein denaturation was calculated using the equation below:

$$\% \text{ Inhibition} = (\text{Absorbance of control} - \text{Absorbance of sample} / \text{Absorbance of control}) \times 100.$$



Figure 3.8.1

3.9 CYTOTOXICITY ASSAY:

Brine Shrimp Lethality Test (BSLT): It is a rapid and inexpensive assay used to screen for general toxicity and potential cytotoxicity of viable plant concentrates that can inhibit cellular proliferation, thus providing a qualitative assessment of its potential anti-neoplastic properties.

To prepare the solution, 2g of NaCl was measured and dissolved in 200 mL of distilled water. The experiment needed an enzyme-linked immunosorbent assay (ELISA) well plate, and 1 mL of the NaCl solution and 1 mL of the flower extracts were deposited into 4 wells located at both the top-right and bottom-left corners. Following this, 10 nauplii were carefully introduced into each subsequent well, with assorted concentrations (100 μ L, 50 μ L, 25 μ L) of *C. guianensis* flower extracts using ethanol, acetone, and chloroform. The fourth well was prompted as a control and did not include any plant extract. The plates were set aside at room temperature for 3-4 days to facilitate the proliferation of nauplii. The assay was monitored three times daily for 3 days to check nauplii activity.

$LC_{50} = \text{Number of dead nauplii} / (\text{Number of inactive nauplii} + \text{Number of active nauplii}) \times 100.$

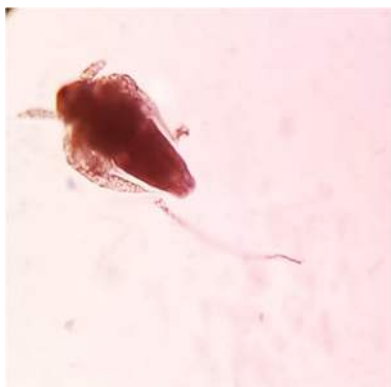


Figure 3.9.1

3.10 MOLECULAR DOCKING:

This study focuses on the chemical compound ($\pm$)-1-tetradecen-3-ol, a long-chain unsaturated alcohol with potential biological and industrial relevance.

It aims to characterise the molecular structure and its optimisation using software applications listed below:

- *Open Babel*
- *Pyrex*
- *Autodock Vina*
- *Protein Data Bank (PDB)*
- *PubChem*

The docking underscores the significance of understanding such compounds for their roles in pheromone signalling, antioxidant activity, and as ligands in molecular modelling.

Key Concepts:

Chemical Structure: ($\pm$)-1-tetradecen-3-ol features a 14-carbon chain with a terminal alkene (C=C at C1) and a hydroxyl group at C3, with the molecule existing as a racemic mixture due to the chiral centre at C3.

Chirality: The presence of a stereocenter makes the molecule exist as enantiomers (R and S), influencing its biological activity.

Antioxidant Potential: Similar compounds, such as phytol, exhibit antioxidant effects by scavenging free radicals, suggesting potential health benefits.

Molecular Structure Acquisition and Optimisation:

Structure Retrieval: 3D structure files (SDF format) were downloaded from chemical databases., PubChem.

Preparation Steps:

- Verified protonation states suitable for physiological pH.
- Generated both enantiomers (R and S) for docking studies.
- Added missing hydrogens and assigned partial charges (Gasteiger or AM1-BCC).
- Defined rotatable bonds, especially in the long alkyl chain and hydroxyl group.
- Saved prepared structures in PDBQT format for docking.

Protein Preparation for Docking:

Target Proteins: Keap1, Protein 3 Inflammasome and NADPH oxidase were selected for their biological relevance.

Structure Retrieval: Downloaded crystal structures from PubChem and RCSB PDB (e.g., Keap1: 4N1B; NADPH oxidase: 5O0X, protein_3_Inflammasome).

Ligand Optimisation:

- Removed water molecules, ions, and co-crystallised ligands.
- Corrected missing residues or side chains using modelling tools.
- Added hydrogens and assigned proper protonation states at pH 7.4.
- Ensured the structures are suitable for accurate molecular docking simulations.

4. RESULTS

Increasing global interest in repurposing natural drug-like phytochemicals over synthetic ones, because many plants have shown promising potential for clinical use through scientific research and evaluation. While indigenous communities have long used plants for medical treatments, often without formal scientific backing, modern studies are now verifying many of these remedies, though some still lack scientific validation. Plants produce secondary metabolites that help defend against pathogens, suggesting their benefits for immune support and the treatment of various diseases.

Pharmacognosy focuses on studying drug substances derived from natural sources, vital for developing and authenticating new medications. This field provides reliable methods to gather comprehensive information about crude drugs. The current research emphasizes the pharmacognostic analysis of *C. guianensis*, contributing to the standardisation and scientific understanding of its medicinal properties.

Phytochemical Analysis of *Couroupita guianensis* using different solvents like ethanol, acetone, and chloroform reveals a varied distribution of secondary metabolites, which can be attributed to the different polarities of the extraction solvents. The overall pattern suggests that ethanol is generally the most effective solvent for extracting a broad spectrum of compounds, especially the more polar ones like flavonoids and tannins. The results clearly demonstrate the principle of "like dissolves like," where the choice of solvent significantly affects the types and amounts of phytochemicals extracted from *C. guianensis*.

Ethanol, a highly polar protic solvent, consistently extracted the greatest diversity and quantity of polar compounds. This includes the sub-class compounds of Flavonoids, Tannins, Saponins, and Phenols. The high polarity allows it to break down the highly polar chemical bonds and hydrogen bonds characteristic of these compounds.

Acetone, a medium-polarity aprotic solvent, shows good efficiency for moderately polar compounds. It typically extracts a considerable volume of flavonoids and tannins, and often performs well for compounds like Terpenoids.

Chloroform, a low-polarity solvent, mainly extracts less polar compounds such as Sterols/Phytosterols and, to a lesser extent, terpenes and alkaloids. Polar compounds like carbohydrates and most flavonoids were typically found in low/trace amounts or were absent.

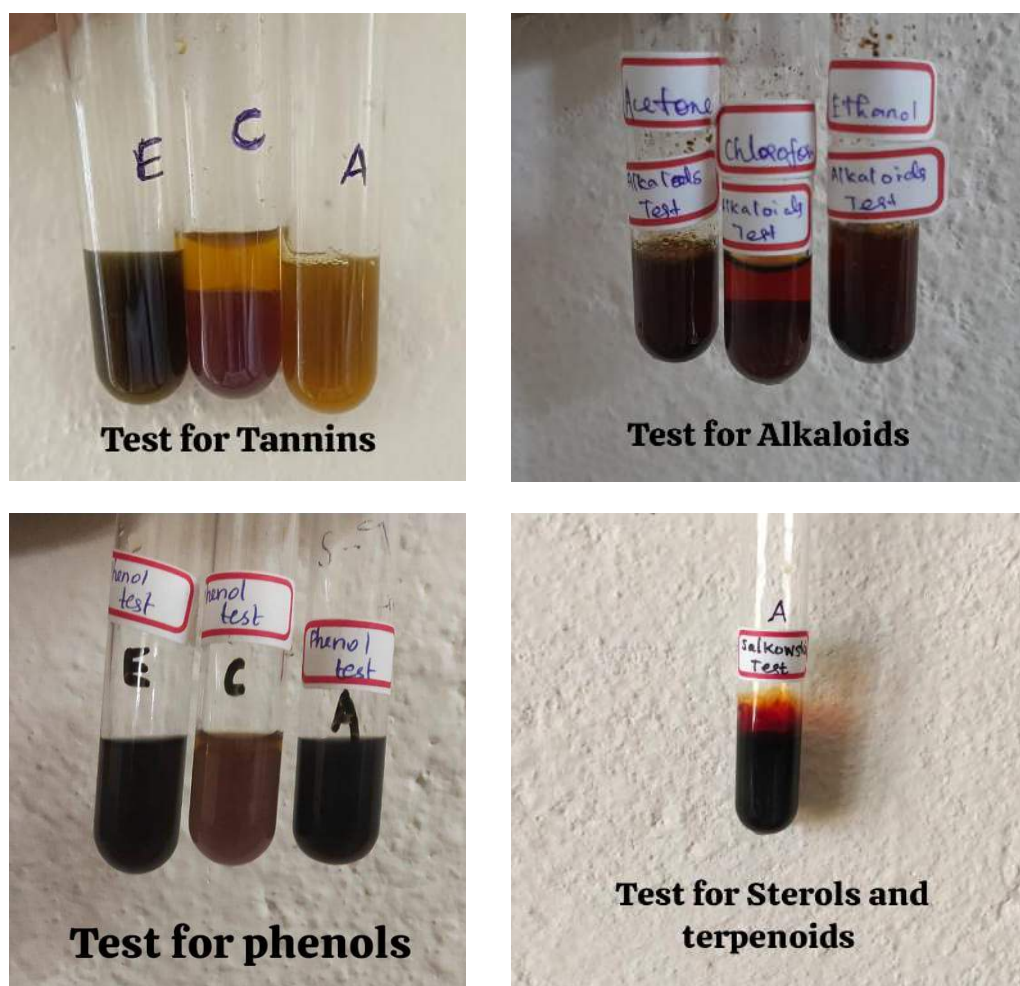


Figure 4.1

The results of phytochemical tests are shown in Figure 4.1 and in Table 4.1, eliciting the activity of extracts using three different organic solvents.

S. NO	TEST	CHLOROFORM	ETHANOL	ACETONE
1.	Alkaloids			
	Wagner's test	+	+	+
2.	Flavonoids			
	Alkaline test	-	+	+
3.	Proteins			
	Biuret test	+	+	+
	Xanthoprotein test	+	+	+
	Concentrated H ₂ SO ₄ test	+	+	+
4.	Saponins			
	Foam test	-	-	-
5.	Tannins			
	HCL test	-	-	-
	Braymer's test	-	-	-
6.	Coumarins	-	+	+
7.	Quinones			
	Alcoholic KOH test	-	-	-
8.	Glycosides	-	+	-
9.	Sterols and Terpenoids			
	Salkowski's test	+	+	+
10.	Reducing sugars			
	Fehling's test	-	-	-
	Benedict's test	-	-	-
11.	Carboxylic acid			
	Sodium Bicarbonate test	-	-	-
12.	Lipids	-	-	+
13.	Phenols			
	Ferric Chloride test	-	+	+
	Lead acetate test	-	+	-
14.	Triterpenoids	-	+	-

Table 4.1

The Gas Chromatography-Mass Spectrometry revealed that the given sample of *Couroupita guianensis* showed six compounds. We chose two peak compounds from the GC-MS analysis of the *Couroupita guianensis* sample, and they are:

✚ (±)-1-Tetradecen-3-ol (C 14H 28 O): An unsaturated long-chain fatty alcohol.

✚ 1-Butyne-3-ethoxy (C 6 H 10O): A volatile ethoxy-alkyne.

Identification of these two compounds provides chemical support for the plant's diverse traditional uses. Functions and applications of identified key compounds:

(±)-1-Tetradecen-3-ol is a fatty alcohol that shows antimicrobial and antioxidant properties, which are relevant to the plant's reported wound-healing and medicinal uses.

1-Butyne-3-ethoxy is a simple, volatile compound that contributes to the extract's less-polar fraction. Volatile components often possess antifungal and aromatic properties, aligning with the traditional use of the plant's fragrance.

The ethanol extract showed both semi-polar fatty derivatives and more volatile components, suggesting a broad spectrum of compounds capable of potential pharmacological activities and is listed below:

Peak	Area%	A/H	Name
1	0.31	1.54	(+)-1-TETRADECEN-3-OL
2	0.48	1.51	Pentane, 1,1-diethoxy-
3	0.86	1.8	BENZOIC ACID, 2,5-BIS(TRIMETHYLSILOXY)-, TRIMETHYLSILYL ESTER
4	3.44	1.88	Cyclohexasiloxane, dodecamethyl-
5	0.58	1.64	1,3-DIPHENYL-1-((TRIMETHYLSILYL)OXY)-1(Z)-HEPTENE

6	0.49	1.94	1,4-PENTADIEN-3-ONE, 1,5-BIS(2-ETHYNYLPHENYL)-
7	6.06	2.44	n-Hexadecanoic acid
8	17.09	2.14	HEXADECANOIC ACID, ETHYL ESTER
9	4.72	1.98	trans, trans-9,12-Octadecadienoic acid, propyl ester
10	0.81	2.19	SYN/ANTI-5,5-DIMETHYL-6- OXOBICYCLO [2.2.2] OCTANE-2-CARBALDEHYDE
11	0.56	1.88	UNDECANOIC ACID, 2-METHYL-, METHYL ESTER
12	0.63	2.18	2-DECYLOXYETHANOL
13	0.2	1.2	1,2-DIMETHOXYETHENE
14	31.72	3.38	13-Docosenamide, (Z)-
15	28.25	2.27	Squalene
16	0.95	5.95	ETHYL (Z)-2-ACETOXY-3- PHENYLAMINOPROPENOATE
17	0.29	1.14	1- (BISTRIFLUOROMETHYLAMINO- OXY)-1-METHYLCYCLOHEXANE
18	0.95	4.86	13,14,15,16,17-PENTANORLABD- 8-EN-12-OL
19	0.35	2.2	2-HEPTANONE, 3,6-DIMETHYL-3-(PHENYLTHIO)-
20	1.25	3.96	ETHANONE, 1,1'-[3,3'-BIISOXAZOLE] -5,5'DIYLBIS-

Table 4.2

Antibacterial Activity of organic samples was evaluated comparatively. The zone of inhibition for each extract was determined by looking at the Well diffused petri plates and recording it using the conventional zone measurement scale. The extracts were evaluated using 4 strains of bacterial pathogens. Ethanol and Acetone showed significantly higher activity against all tested microbes. Chloroform showed the least activity among the chosen solvents.

S. No	Bacterial Strains	Chloroform				
		Control 1	Control 2	C1	C2	C3
1.	<i>Streptococcus Pneumonia</i>	0.5cm	2.5cm	0.4cm	0.3cm	1.4cm
2.	<i>Pseudomonas aeruginosa</i>	0.6cm	2.5cm	0.6cm	0.6cm	1.6cm
3.	<i>Klebsiella pneumonia</i>	1.0cm	2.7cm	0.3cm	0.9cm	1.8cm
4.	<i>Staphylococcus aureus</i>	0.7cm	3.0cm	0.4cm	1.0cm	1.9cm

Table 4.3

S. No	Bacterial Strains	Acetone				
		Control 1	Control 2	A1	A2	A3
1.	<i>Streptococcus Pneumonia</i>	0.5cm	2.3cm	0.5cm	1.5cm	2.0cm
2.	<i>Pseudomonas aeruginosa</i>	1.0cm	3.0cm	0.6cm	1.4cm	2.2cm
3.	<i>Klebsiella pneumonia</i>	0.8cm	2.8cm	0.8cm	1.6cm	2.3cm
4.	<i>Staphylococcus aureus</i>	1.1cm	2.6cm	0.9cm	1.2cm	2.0cm

Table 4.4

S. No	Bacterial Strains	Ethanol				
		Control 1	Control 2	E1	E2	E3
1.	<i>Streptococcus Pneumonia</i>	0.4cm	2.7cm	1.0cm	1.5cm	2.1cm
2.	<i>Pseudomonas aeruginosa</i>	1.0cm	2.6cm	1.2cm	1.4cm	2.3cm
3.	<i>Klebsiella pneumonia</i>	0.9cm	2.8cm	1.0cm	1.8cm	2.0cm
4.	<i>Staphylococcus aureus</i>	0.5cm	2.4cm	0.8cm	1.7cm	1.9cm

Table 4.5

Anti-inflammatory Assay revealed the aqueous extract's anti-inflammatory potential using the bovine serum albumin denaturation assay by inspecting their inhibitory activity at 280nm through a UV-Vis spectrophotometer. At 100 µg/mL, it achieved 74% inhibition, showing notable effectiveness in preventing protein denaturation. When compared to Aspirin across all concentrations, the extract demonstrated promising anti-inflammatory and possible antioxidant activity.

Sample (µg/mL)	% of Inhibition
Sample A	60.6%
Sample B	69.6%
Sample C	74.5%
Positive control	97.8%

Table 4.6

Sample (µg/mL)	OD value (at 280nm)
Blank 1	0.0023
Blank 2	0.0006
Sample A	1.13
Sample B	0.73
Sample C	0.87
Positive Control	2.87
Negative Control	0.2086

Table 4.7

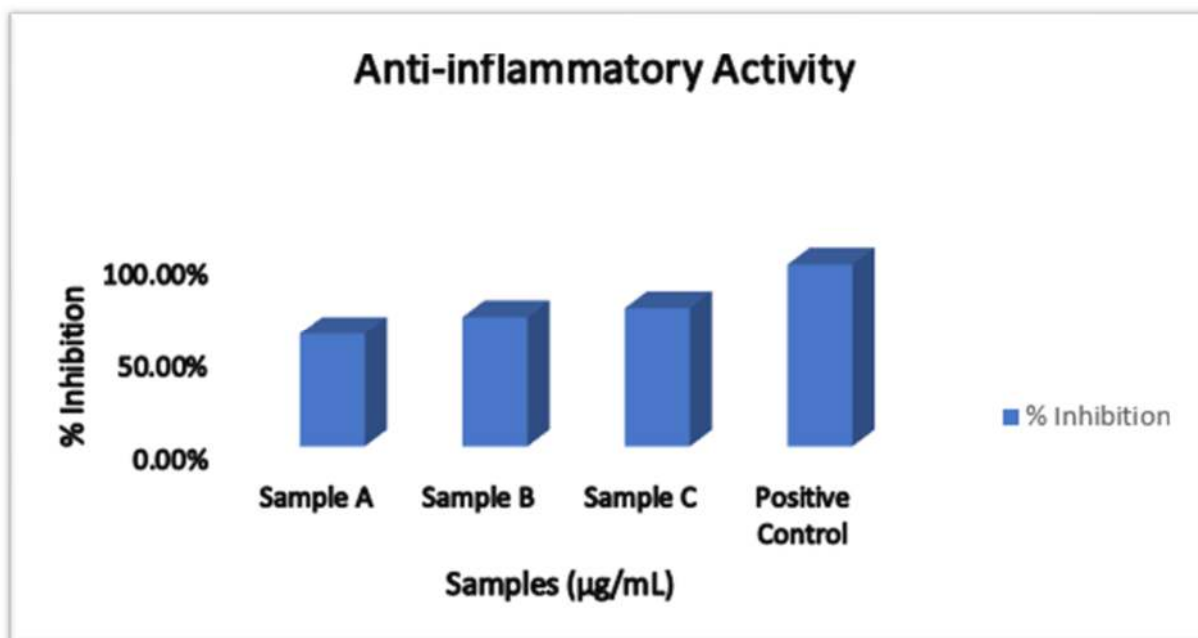


Figure 4.2

Cytotoxicity Assay of both organic and aqueous extracts were examined through the brine shrimp lethality bioassay. This methodology serves as a standard approach for determining cytotoxic potential by assessing the lethal effects on brine shrimp nauplii populations. During this investigation, a reference group containing only saline solution was established to provide a comparison standard for determining the survival percentage of living nauplii. This evaluation demonstrated that various concentrations of organic and aqueous preparations produced distinct impacts on nauplii viability. When exposed to organic extracts at a 25µg/mL concentration, roughly 70% of organisms remained viable. Correspondingly, exposure to 50µg/mL and 100µg/mL concentrations of organic preparations led to the survival of approximately 45% and 25% of nauplii populations, respectively. These findings demonstrate increased cytotoxic potential that qualitatively corresponds with anti-cancer properties.

Sample (Aqueous Extract)	Nauplii Activity (LC50)
Q1 (25µg/mL)	70%
Q2 (50µg/mL)	45%
Q3 (100µg/mL)	25%
Negative Control	90%

Table 4.8

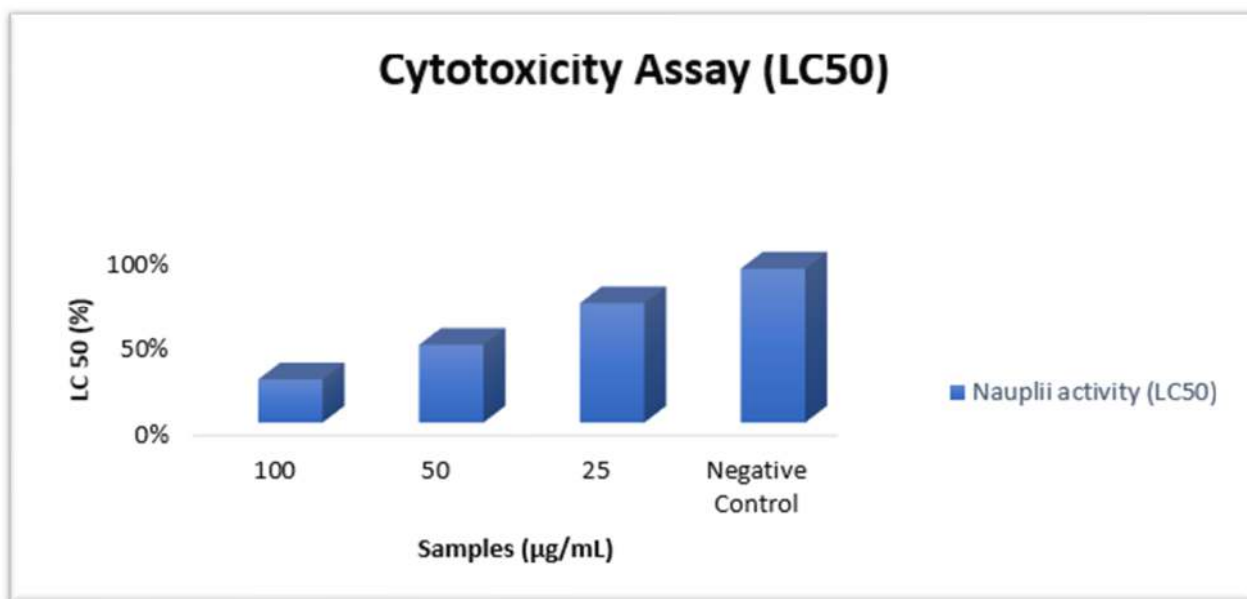


Figure 4.3

Molecular Docking studies revealed that the protein_3_inflammasome ligand has binding affinities ranging from -2.6 to -2.3 kcal/mol, with the strongest binding seen at -2.6. Several docking poses show varied RMSD values, with the best-scoring pose (affinity -2.6) displaying no deviation (RMSD 0.0), suggesting a highly stable interaction in the top pose.

Key Findings:

- The ($\pm$)-1-tetradecen-3-ol molecule exists as a racemic mixture with distinct enantiomers influencing binding affinity.
- Structural optimisation confirmed the molecule's flexibility due to the long hydrocarbon chain and hydroxyl group.
- Docking results indicated differential binding energies between enantiomers, suggesting that stereochemistry impacts biological interactions.
- The compound's structural features align with known pheromonal and antioxidant activities observed in related natural products.

A	B	C	D
Ligand	Binding Affinity	rmsd/ub	rmsd/lb
protein_3_inflammasome_ligand.	-2.6	0	0
protein_3_inflammasome_ligand.	-2.5	2.681	1.35
protein_3_inflammasome_ligand.	-2.4	2.609	1.437
protein_3_inflammasome_ligand.	-2.4	4.289	1.638
protein_3_inflammasome_ligand.	-2.3	2.583	1.505
protein_3_inflammasome_ligand.	-2.3	2.321	1.524
protein_3_inflammasome_ligand.	-2.3	5.418	2.473
protein_3_inflammasome_ligand.	-2.3	3.368	1.922
protein_3_inflammasome_ligand.	-2.3	5.931	2.464

Figure 3.10.1

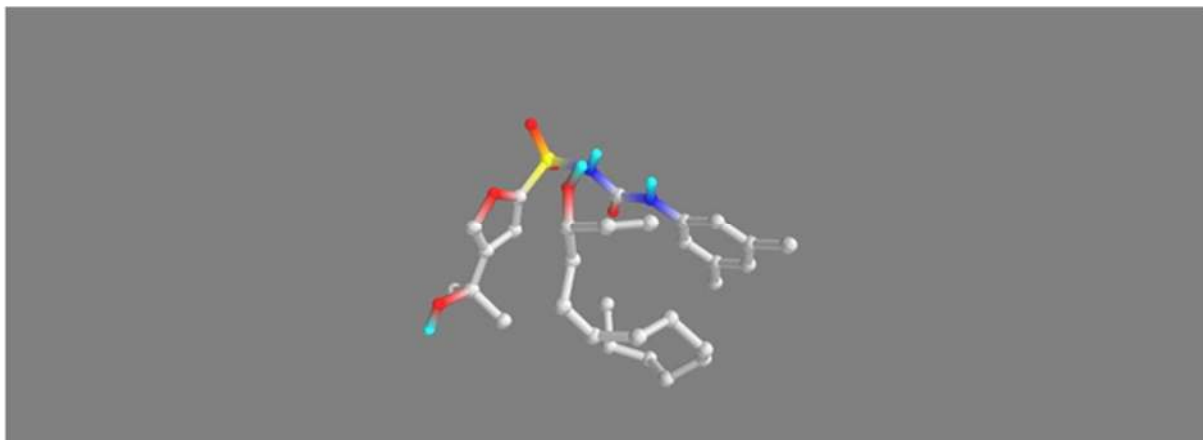


Figure 3.10.2

5. DISCUSSION:

Identifying novel pharmacological solutions from natural origins for resistant bacteria is an urgent challenge because of their significant effects, and establishing evidence-based herbal treatments seems crucial for advancing innovative therapies, combating antimicrobial resistance, minimising side effects, and utilising the extensive phytochemical variety of natural substances. This effort offers potential benefits for enhancing patient outcomes and highlights the importance of merging millennia-old traditional wisdom with contemporary scientific investigation.

Phytochemical analysis using ethanol, acetone, and chloroform as solvents reveals varying concentrations of secondary metabolites, with ethanol proving most effective for extracting a wide array of polar compounds, including flavonoids and tannins. This is consistent with the concept of "like dissolves like," since ethanol is very polar and can efficiently disrupt the polar chemical bonds within these compounds. Acetone, possessing medium polarity, shows good extraction capabilities for moderately polar compounds, while chloroform, with low polarity, mainly extracts less polar compounds such as sterols and alkaloids.

Gas Chromatography-Mass Spectrometry (GC-MS) analysis of the ethanol extract has identified six volatile and semi-volatile compounds. Notable among these is ($\pm$)-1-Tetradecen-3-ol, known for antimicrobial and antioxidant properties linked to wound healing, and 1-Butyne-3-ethoxy, a volatile compound associated with antifungal and aromatic properties.

Antibacterial evaluations of the organic extracts from *C. guianensis* flowers indicate that both ethanol and acetone extracts exhibit significant antibacterial activity against various microbial strains, while chloroform extracts show minimal effectiveness. This study enhances the understanding of the medicinal potential of *C. guianensis* and underscores the importance of investigating plant-based pharmacological resources. Antifungal activity was assessed against four strains from different fungal families, showing no significant effects across three concentrations.

Anti-inflammatory activity of the aqueous extract was evaluated using the bovine serum albumin denaturation assay, revealing a maximum inhibition of 47% at 100µg/mL. This indicates notable anti-inflammatory properties, particularly when compared to the positive control, Aspirin. Cytotoxicity of both organic and aqueous extracts was tested using the brine shrimp lethality assay, where varying concentrations impacted the survival rates of nauplii. At 25µg/mL of organic extracts, about 70% survived, whereas survival was 45% at 50µg/mL and 25% at 100µg/mL, suggesting a correlation between increased cytotoxicity and anti-neoplastic activity. Overall, our study highlights the ability of herbal formulation extraction to effectively assess, isolate and develop peak compounds in organic flower extracts of *C. guianensis*.

CONCLUSION:

The ongoing global challenge of antimicrobial resistance and the demand for safer, naturally derived pharmaceuticals have intensified the search for bioactive compounds in medicinal plants. Among these, the Cannonball Tree, *Couroupita guianensis* Aubl. (Family: Lecythidaceae), stands out as a species widely recognised in traditional medicine systems across South Asia and South America, particularly the inflorescence, which has been recurrently used to treat ailments ranging from skin diseases and pain to inflammatory conditions and microbial infections. Phytochemical analysis confirmed the presence of key secondary metabolites, including alkaloids and flavonoids, with a strong correlation to biological activities, especially Sterols and Triterpenes. Gas Chromatography-Mass Spectrometry identified specific compounds, supporting the extract's traditional use. Molecular docking studies showed that (±)-1-tetradecen-3-ol has properties promoting biological activity, suggesting potential interactions with proteins relevant to oxidative stress. The flower extract exhibited antimicrobial effects and better activity against protein denaturation, confirming its traditional use for inflammation. The Brine Shrimp Lethality Test indicated the extract's cytotoxicity, suggesting potential anti-neoplastic activity. This comparative analysis of the *C. guianensis* flower extract successfully integrates phytochemical screening, GC-MS characterization, and a suite of biological assays.

REFERENCES:

1. Albuquerque, B. R., Heleno, S. A., Oliveira, M. B. P., Barros, L., & Ferreira, I. C. (2021). Phenolic compounds: Current industrial applications, limitations and future challenges. *Food & function*, 12(1), 14-29.
2. Bennett, B.C., & Husby, C.E. (2008). Patterns of medicinal plant use: an examination of the Ecuadorian Shuar medicinal flora using contingency table and binomial analyses. *Journal of Ethnopharmacology*, 116 (3), 422-30.
3. De Albuquerque, U. P. (2006). Re-examining hypotheses concerning the use and knowledge of medicinal plants: a study in the Caatinga vegetation of NE Brazil. *Journal of ethnobiology and ethnomedicine*, 2(1), 30.
4. de Medeiros, P. M., Ladio, A. H., & Albuquerque, U. P. (2013). Patterns of medicinal plant use by inhabitants of Brazilian urban and rural areas: a macroscale investigation based on available literature. *Journal of ethnopharmacology*, 150(2), 729-746.
5. Esposito, T., Pisanti, S., Martinelli, R., Celano, R., Mencherini, T., Re, T., & Aquino, R. P. (2023). Couroupita guianensis bark decoction: From Amazonian medicine to the UHPLC-HRMS chemical profile and its role in inflammation processes and re-epithelialization. *Journal of Ethnopharmacology*, 313, 116579.
6. Gousia, S. K., Kumar, K. A., Kumar, T. V., & Latha, J. N. L. (2013). Biological activities and medicinal properties of Couroupita guianensis. *International Journal of Pharmacy and Pharmaceutical Science Research*, 3(4), 140-143.
7. Jagannatha, S., Maity, S., Santra, A., Kuruvalli, G., Reddy, V.D., & Golla, R. (2025). An Updated Review on Ethnomedicinal Uses, Phytochemistry and Pharmacological Activities of Couroupita guianensis. *Pharmacognosy Reviews*.
8. Kaneria, M.J., Parmar, J.B., & Rakholiya, K.D. (2019). Molecular docking and drug design of phytoconstituents from Couroupita guianensis – An in-silico perspective. *Journal of Pharmacognosy and Phytochemistry*, 8, 53-60.
9. Kekuda, T. P. (2017). Antimicrobial, radical scavenging, and insecticidal activity of leaf and flower extracts of Couroupita guianensis Aubl. *International Journal of Green Pharmacy (IJGP)*, 11(03).
10. Kumar, C.S., Naresh, G., Sudheer, V., Veldi, N., & Elumalai, A. (2011). A SHORT REVIEW ON THERAPEUTIC USES OF COUROUPITA GUIANENSIS AUBL.
11. Leandro de França Ferreira, É., Pereira de Carvalho Oliveira, J., Silva de Araújo, M.R., Rai, M., & Chaves, M.H. (2021). Phytochemical Profile and

Ethnopharmacological Applications of Lecythidaceae: An overview. *Journal of ethnopharmacology*, 114049.

12. McCulley, S.F., & Setzer, W.N. (2014). An in-silico investigation of anti-Chagas phytochemicals. *Current clinical pharmacology*, 9 3, 205-57.
13. Ogungbe, I.V., Erwin, W.R., & Setzer, W.N. (2014). Antileishmanial phytochemical phenolics: molecular docking to potential protein targets. *Journal of molecular graphics & modelling*, 48, 105-17 .
14. Pandurangan, P., Sahadeven, M., Sunkar, S., & Dhana, S. K. N. M. (2018). Comparative analysis of biochemical compounds of leaf, flower and fruit of *Couroupita guianensis* and synthesis of silver nanoparticles. *Pharmacognosy Journal*, 10(2).
15. Pinheiro, M.M., Fernandes, S.B., Fingolo, C.E., Boylan, F., & Fernandes, P.D. (2013). Anti-inflammatory activity of ethanol extract and fractions from *Couroupita guianensis* Aublet leaves. *Journal of ethnopharmacology*, 146 1, 324-30 .
16. Pusapati, M. R., Guntuku, G., Yarlagadda, A. C., Nagaraju, G., Soumya, M., & Lakshmi, T. B. V. (2014). An Overview of Classification, Production and Biomedical Applications of Biosurfactants. *Research Journal of Pharmacy and Technology*, 7(5), 608.
17. Ramalakshmi, C.S., Ranjitsingh, A.J., Kalirajan, K., Kalirajan, A., Athinarayanan, G., & Mariselvam, R. (2013). A Preliminary screening of the Medicinal Plant *Couroupita guianensis* for its Antimicrobial Potential against Clinical and Fish-borne pathogens.
18. Sheba, L. A., & Anuradha, V. (2020). An updated review on *Couroupita guianensis* Aubl: a sacred plant of India with myriad medicinal properties. *J Herbmed Pharmacol*, 9(1), 1-11.
19. Singh, R., Kumari, N., Gangwar, M., & Nath, G. (2015). QUALITATIVE CHARACTERIZATION OF PHYTOCHEMICALS AND IN VITRO ANTIMICROBIAL EVALUATION OF LEAF EXTRACT OF COUROUPITA GUIANENSIS AUBL. - A THREATENED MEDICINAL TREE. *International Journal of Pharmacy and Pharmaceutical Sciences*, 7, 212-215.
20. Sundararajan, R., & Koduru, R. (2014). A complete profile on *Couroupita guianensis*-traditional uses, pharmacological activities and phytoconstituents. *Pharmacophore*, 5(1-2014), 147-159.
21. Swapnalatha, S., & Rajeswari, V. D. (2014). Antidiabetic activity of *couroupita guianensis*—a review. *IOSR Journal of Pharmacy and Biological Sciences*, 9(3), 41-43.

22. Trotter, R. T., & Logan, M. H. (2019). Informant consensus: a new approach for identifying potentially effective medicinal plants. In *Plants and indigenous medicine and diet* (pp. 91-112). Routledge.
23. Ubani, A., Agwom, F., Shehu, N. Y., Luka, P., Umera, A., Umar, U., ... & Aguiyi, J. C. (2020). Molecular docking analysis of some phytochemicals on two SARS-CoV-2 targets. *BioRxiv*, 2020-03.
24. Wassén, S.H., & Holmstedt, B.R. (1963). The use of paricá, an ethnological and pharmacological review. *Ethnos*, 28, 5-45.